

MPI_Init(&argc, &argv).

int MPI_Comm_rank(MPI_Comm comm, int *rank);

int MPI_Comm_size(MPI_Comm comm, int *size);

**int MPI_Send(void *buf, int count, MPI_Datatype dtype, int dest, int tag,
MPI_Comm comm);**

**int MPI_Recv(void *buf, int count, MPI_Datatype dtype, int source, int tag,
MPI_Comm comm, MPI_Status *status);**

**int MPI_Isend(void *buf, int count, MPI_Datatype dtype, int dest, int tag,
MPI_Comm comm, MPI_Request *request);**

**int MPI_Irecv(void *buf, int count, MPI_Datatype dtype, int source, int tag,
MPI_Comm comm, MPI_Request *request);**

int MPI_Wait(MPI_Request *request, MPI_Status *status);

int MPI_Test(MPI_Request *request, int *flag, MPI_Status *status);

int MPI_Barrier (MPI_Comm comm)

**int MPI_Reduce (void* send_buffer, void* recv_buffer, int count, MPI_Datatype
datatype, MPI_Op operation, int rank, MPI_Comm comm)**

**int MPI_Scan(void* send_buffer, void* recv_buffer, int count, MPI_Datatype
datatype, MPI_Op operation, MPI_Comm comm)**

**int MPI_Bcast (void* buffer, int count, MPI_Datatype datatype, int rank,
MPI_Comm comm)**

**int MPI_Scatter (void* send_buffer, int send_count, MPI_datatype send_type,
void* recv_buffer, int recv_count, MPI_Datatype recv_type, int rank, MPI_Comm
comm)**

**int MPI_Gather (void* send_buffer, int send_count, MPI_datatype send_type,
void* recv_buffer, int recv_count, MPI_Datatype recv_type, int rank, MPI_Comm
comm)**

**int MPI_Type_struct(int count, int *array_of_blocklengths, MPI_Aint
*array_of_displacements, MPI_Datatype *array_of_types, MPI_Datatype
*newtype)**

MPI_Address(void *location, MPI_Aint * adress)

int MPI_Type_contiguous(int count, MPI_Datatype oldtype, MPI_Datatype *newtype)

int MPI_Type_vector(int count, int blocklength, int stride, MPI_Datatype oldtype, MPI_Datatype *newtype)

int MPI_Type_indexed(int count, int *array_of_blocklengths, int *array_of_displacements, MPI_Datatype oldtype, MPI_Datatype *newtype)

int MPI_Type_create_subarray(int ndims, int *sizes, int *subsizes, int *offsets, int order, MPI_Datatype oldtype, MPI_Datatype *newtype)

int MPI_Type_commit (MPI_datatype *datatype)

int MPI_Type_create_resized(MPI_Datatype oldtype, MPI_Aint lb, MPI_Aint extent, MPI_Datatype *newtype)

int MPI_Comm_group(MPI_Comm comm, MPI_Group *group)

int MPI_Group_rank(MPI_Group group, int *rank)

int MPI_Group_size(MPI_Group group, int *size)

int MPI_Group_excl(MPI_Group group, int count, int *nonmembers, MPI_Group *new_group)

int MPI_Group_incl(MPI_Group old_group, int count, int *members, MPI_Group *new_group)

int MPI_Group_intersection(MPI_Group group1, MPI_Group group2, MPI_Group *newgroup)

int MPI_Group_union(MPI_Group group1, MPI_Group group2, MPI_Group *newgroup)

int MPI_Group_difference(MPI_Group group1, MPI_Group group2, MPI_Group *newgroup)

int MPI_Comm_create(MPI_Comm old_comm, MPI_Group, MPI_Comm *new_comm)

int MPI_Comm_split(MPI_Comm, int color, int key, MPI_Comm *new_comm)

int MPI_Cart_create(MPI_Comm old_comm, int ndims, int *dim_size, int *periods, int reorder, MPI_Comm *new_comm)

int MPI_Cart_coords(MPI_Comm comm, int rank, int maxdims, int *coords)

int MPI_Cart_rank (MPI_Comm comm, int *coords, int *rank)

int MPI_Cart_shift(MPI_Comm comm, int direction, int disp, int *rank_source, int *rank_dest)

int MPI_Sendrecv(const void *sendbuf, int sendcount, MPI_Datatype sendtype, int dest, int sendtag, void *recvbuf, int recvcount, MPI_Datatype recvtpe, int source, int recvtag, MPI_Comm comm, MPI_Status *status)

int MPI_Sendrecv_replace(void *buf, int count, MPI_Datatype datatype, int dest, int sendtag, int source, int recvtag, MPI_Comm comm, MPI_Status * status)

----- MPI I/O

int MPI_File_open(MPI_Comm comm, const char *filename, int amode, MPI_Info info, MPI_File *fh)

int MPI_File_seek(MPI_File fh, MPI_Offset offset, int whence)

int MPI_File_read(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_write(MPI_File fh, const void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_close(MPI_File *fh)

int MPI_File_read_at(MPI_File fh, MPI_Offset offset, void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_write_at(MPI_File fh, MPI_Offset offset, const void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_set_view(MPI_File fh, MPI_Offset disp, MPI_Datatype etype, MPI_Datatype filetype, const char * datarep, MPI_Info info)

int MPI_File_read_all(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_write_all(MPI_File fh, const void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_Type_create_darray(int size, int rank, int ndims, int array_of_gsizes[], int array_of_distribs[], int array_of_dargs[], int array_of_psize[], int order, MPI_Datatype oldtype, MPI_Datatype *newtype)

int MPI_File_iread(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Request* request)

int MPI_File_iwrite(MPI_File fh, ROMIO_CONST void *buf, int count, MPI_Datatype datatype, MPI_Request* request)

int MPI_File_iwrite_at(MPI_File fh, MPI_Offset offset, const void *buf, int count, MPI_Datatype datatype, MPI_Request* request);

int MPI_File_iread_at(MPI_File fh, MPI_Offset offset, void *buf, int count, MPI_Datatype datatype, MPI_Request* request)

int MPI_File_write_all_begin(MPI_File fh, const void *buf, int count, MPI_Datatype datatype);

int MPI_File_write_all_end(MPI_File fh, const void *buf, MPI_Status *status);

int MPI_File_write_shared(MPI_File fh, const void *buf, int count, MPI_Datatype datatype, MPI_Status *status);

int MPI_File_read_shared(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Status * status);

int MPI_File_seek_shared(MPI_File fh, MPI_Offset offset, int whence);

int MPI_File_iwrite_shared(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Request *request)

int MPI_File_iread_shared(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Request *request)

int MPI_File_read_ordered(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_write_ordered(MPI_File fh, void *buf, int count, MPI_Datatype datatype, MPI_Status *status)

int MPI_File_set_atomicity (MPI_File fh, int flag)

int MPI_File_sync (MPI_File fh)

int MPI_File_get_size(MPI_File fh, MPI_Offset * size)